Java Classes and Objects

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Java Class...

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In object oriented programming classes and objects play a vital role in programming. These are the two main pillars of OOP(Object Oriented Programming). Without classes and objects, we cannot create a program in Java.

⇒ local variable
⇒ instance variable
⇒ static or class variable.

What is a Class in Java?
A class is a group of objects that have common properties. It is a template or blueprint from which objects are created. It is a logical entity. It cannot be physical.

A Java class contains:

1. **Fields:** Variables stated inside a class that indicate the status of objects formed from that class are called fields, sometimes referred to as instance variables. They specify the data that will be stored in each class object. Different access modifiers, such as public, private and protected, can be applied to fields to regulate their visibility and usability.

2. **Methods:** Methods are functions defined inside a class that include the action or behavior that object of the class are capable of performing. These are techniques allow the outside world to function and change the object state or fields. Additionally, method can be void (that is, they return nothing) or have different access modifiers. They can also return values.

3. **Constructors:** Constructors are unique methods that are used to initialize class objects. When an object of a class is created using the new keyword, they are called with the same name as the class. Constructor can initialize the fields of an object or carry out any additional setup that is required when an object is created.

```
public class Student{
    int rollNum;
    String name;
```

instance variable

```
    Student(int rn, String nm)
    {
        local variable
        rollNum = rn;
        Name = nm;
    }
    ...
}
```

Java class Syntax

```
public class class_name{
    Field;
    Method;
}
```

What is an object in Java?
An object is a real world entity that has state and behavior in other. words an object is a tangible thing that can be touch and feel like car or chair, pen, laptop etc. The banking system is an example of an intangible object. Every object has a distinct identity, which is usually implemented by a unique ID that the JVM uses internally for identification.

Characteristics of an Object

- **State:** It represents the data or value of an object.
- **Behavior:** It represents the behavior or functionality of an object such as deposit withdrawal, etc.
- **Identity:** An object identity is typically implemented via a unique ID. The IDS value is not visible to the external user. However, it is internally used by the JVM to identify each object uniquely.

  For example, a pen is an object. It's name is Reynolds. Its color is white known as its state. It is used to write. So writing is its behavior.

An object is an instance of a class. A class is a template or blueprint from which objects are created. Therefore, a class is called an object factory. So object is the instance or result of a class.

Object Definitions

- An object is a real world entity.

- An object is a runtime entity.
- The object is an entity that has a state and behavior.
- The object is an instance of a class.

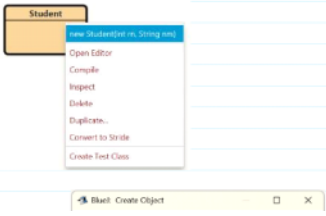
Syntax of an Object

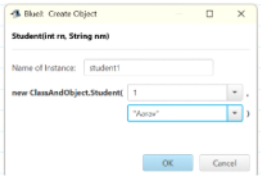
Keyword. optional arguments.

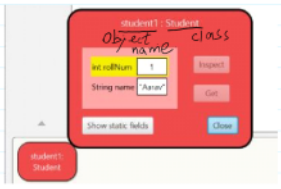
<class_name> variable = new <class_name>(<argument_list>);

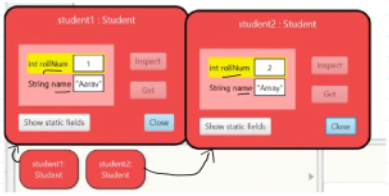
"new" Keyword in Java
The new keyword is used to allocate memory at runtime. All objects get memory in the heap memory area.
In Java, an instance of a class (also referred to as an object) is created. using the new keyword. the new keyword dynamically allocates memory for an object of that class and returns a reference to it when it is followed by the class name and parentheses with optional arguments.

First method to create object in BlueJ graphically.









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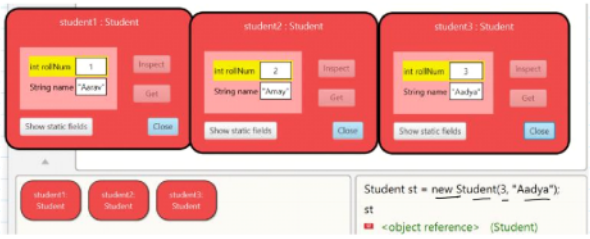
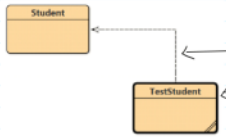




This class has created object of student class.
This property is called "HAS-A" property.
TestStudent "HAS-A" Student
relationship.

```
Blue): Terminal Window - JavaClassIX
Options

Enter roll number: 1
Enter name: Aarav

Content: ClassAndObject.Student@20e286a2
```

package name class name Object ID.

```
Blue): Terminal Window - JavaClassIX
Options

Enter roll number: 1
Enter name: Aarav

Content: ClassAndObject.Student@2583a3ab
Roll Number: 1
Name: Aarav
```

Object ID

hexa-decimal value

RAM

Random Access Memory

```
package ClassAndObject;


/**
 * Write a description of class Student here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Student
{
   //instance variables or fields
   int rollNum;
   String name;
```

```
    //instance variables or fields
    int rollNum;
    String name;

    //Constructor
    Student(int rn, String nm){
      rollNum = rn;
      name = nm;
    }

    //Getter and Setter methods

    int getRollNum(){
      return rollNum;
    }

    String getName(){
      return name;
    }

    void setRollNum(int rn){
      rollNum = rn;
    }

    void setName(String nm){
      name = nm;
    }
  }

  package ClassAndObject;
  import java.util.*;


  /**
   * Write a description of class TestStudent here.
   *
   * @author (your name)
   * @version (a version number or a date)
   */
  public class TestStudent
  {
    public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int rn;
      String nm;
      System.out.print("\nEnter roll number: ");
      rn = sc.nextInt();
      sc.nextLine(); //clears the keyboard buffer
      System.out.print("Enter name: ");
      nm = sc.nextLine();
      Student st1 = new Student(rn, nm);
      System.out.print("\nContent: " + st1);
      System.out.print("\nRoll Number: " + st1.getRollNum());
      System.out.print("\nName: " + st1.getName());
    }
  }
```

local variable

local variable

instance variable

local variable

keyword used to allocate memory

Student st1 = new Student(rn, nm);

class name

constructor

reference variable keeping the reference or address of Object.

New Section 1 Page 4

IDE

code editor + compiler + runtime environment

```
package ClassAndObjects;

/**
 * Write a description of class Account here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Account
{
    //Fields or instance variables
    int acc_num;
    String name;
    float amount;

    //Constructor
    Account(int anum, String nm, float amt){
```

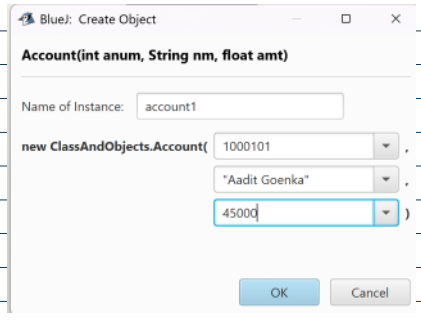
```
        acc_num = anum;
        name = nm;
        amount = amt;
    }

    //Getter and Setter methods
    /**
     * @return the acc_num
     */
    public int getAcc_num() {
        return acc_num;
    }

    /**
     * @param acc_num the acc_num to set
     */
    public void setAcc_num(int acc_num) {
        this.acc_num = acc_num;
    }

    /**
     * @return the name
     */
    public String getName() {
        return name;
    }

    /**
     * @param name the name to set
     */
    public void setName(String name) {
        this.name = name;
    }

    /**
     * @return the amount
     */
    public float getAmount() {
        return amount;
    }

    /**
     * @param amount the amount to set
     */
    public void setAmount(float amount) {
        this.amount = amount;
    }
}
```

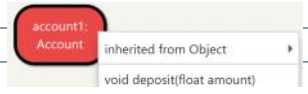
"this" keyword in Java, used to differentiate fields or instance variable with local variable or local parameter, when name of the variables are same.

local variable / parameter

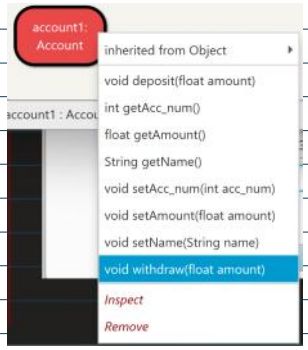
field / instance variable

* "this" keyword holds the reference of current object.
Whatever change we perform it will happen in the current object. The state of current object will update or change.





← When 1st Object or 1st instance is created all the methods of the class gets uploaded on the memory on a separate area alloted for methods.



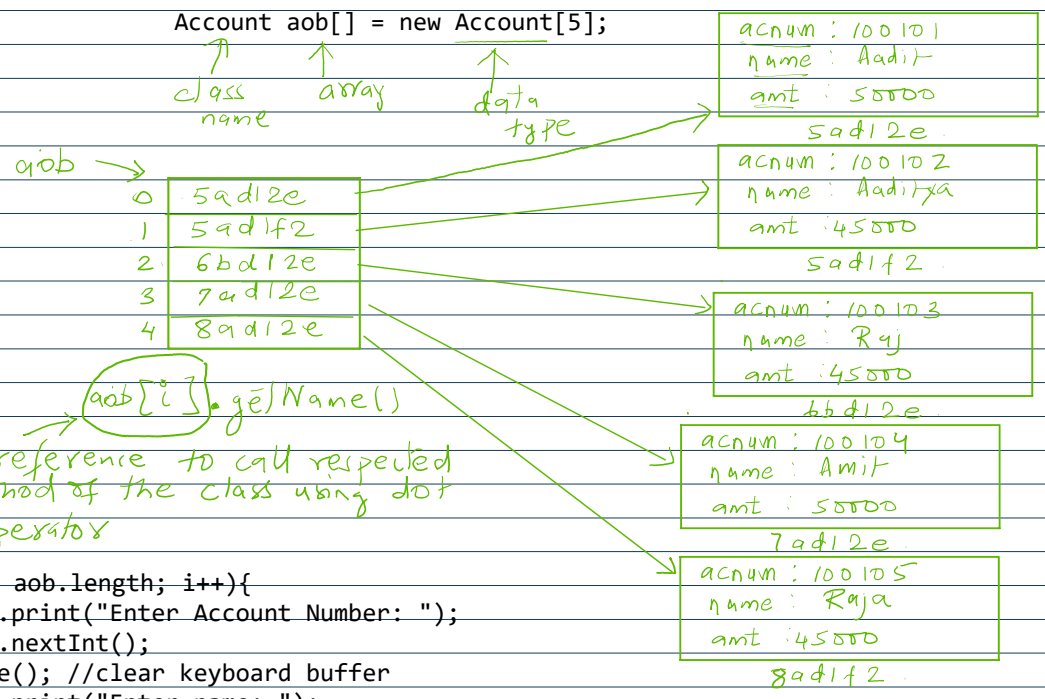
← When 1st Object or 1st instance is created all the methods of the class gets uploaded on the memory on a separate area, alloted for methods.

```
package ClassAndObjects;
import java.util.*;


/**
 * Write a description of class TestAccount here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class TestAccount
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int i, acnum;
        String name;
        float amt;
        System.out.print("\f");
        Account aob[] = new Account[5]; //it will create 5 Account reference variable
        //now we have to allocate memory for the objects.
        for(i = 0; i < aob.length; i++){
            System.out.print("Enter Account Number: ");
            acnum = sc.nextInt();
            sc.nextLine(); //clear keyboard buffer
            System.out.print("Enter name: ");
            name = sc.nextLine();
            System.out.print("Account Balance: ");
            amt = sc.nextFloat();
            aob[i] = new Account(acnum, name, amt); //allocating memory for the object
        }

        System.out.print("\nAccount details of Customers of Jan Dhan Bank");
        for(i = 0; i < aob.length; i++){
            System.out.print("\nAccount number: " + aob[i].getAcc_num());
            System.out.print("\nName: " + aob[i].getName());
            System.out.print("\nBalance: " + aob[i].getAmount());
        }
    }
}

Output:
Enter Account Number: 100101
Enter name: Aadit
Account Balance: 50000
Enter Account Number: 100102
Enter name: Harsh
Account Balance: 40000
Enter Account Number: 100103
Enter name: Faisal
Account Balance: 35000
Enter Account Number: 100104
```

```
Enter name: Raj
Account Balance: 45000
Enter Account Number: 100105
Enter name: Amy
Account Balance: 48000

Account details of Customers of Jan Dhan Bank
Account number: 100101
Name: Aadit
Balance: 50000.0
Account number: 100102
Name: Harsh
Balance: 40000.0
Account number: 100103
Name: Faisal
Balance: 35000.0
Account number: 100104
Name: Raj
Balance: 45000.0
Account number: 100105
Name: Amy
Balance: 48000.0
```

```
Account aob[] = new Account[5];
```

class name — array — data type

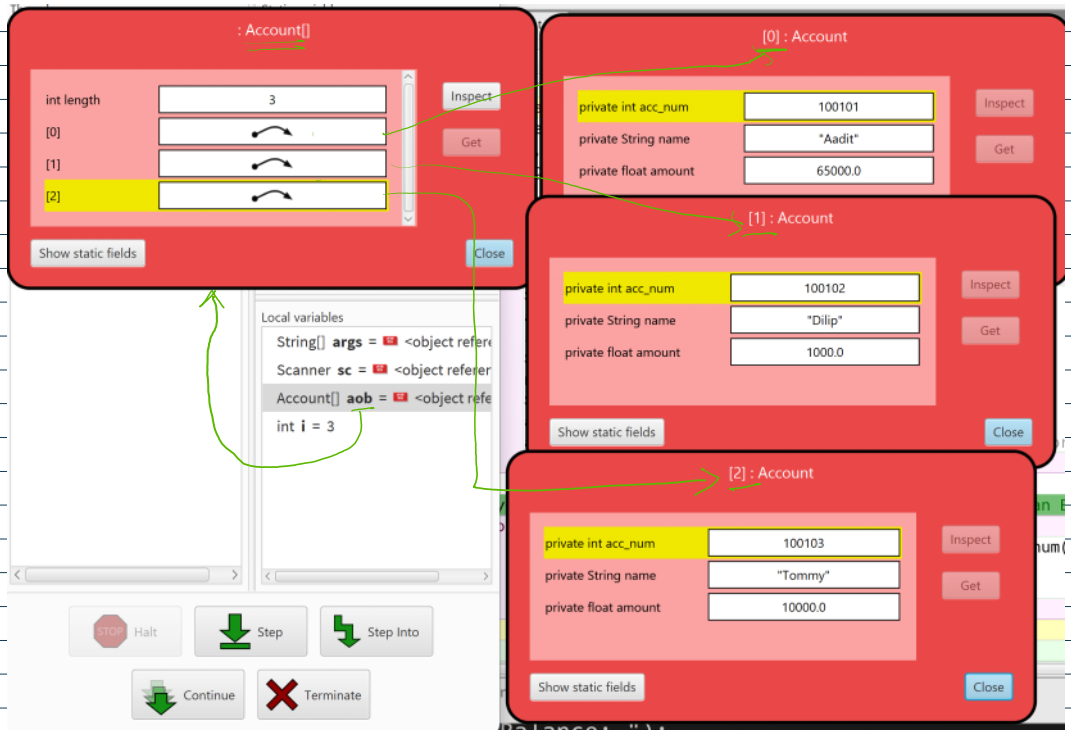


aob[i].getName()

object reference to call respected method of the class using dot operator

```
for(i = 0; i < aob.length; i++){
    System.out.print("Enter Account Number: ");
    acnum = sc.nextInt();
    sc.nextLine(); //clear keyboard buffer
    System.out.print("Enter name: ");
    name = sc.nextLine();
    System.out.print("Account Balance: ");
    amt = sc.nextFloat();
    aob[i] = new Account(acnum, name, amt); //allocating memory for the object
}
```

allocating memory for the object.


: Account[]
int length
3
[0]
[1]
[2]
Inspect
Get
Show static fields
Close
[0] : Account
private int acc_num
100101
private String name
"Aadit"
private float amount
65000.0
[1] : Account
private int acc_num
100102
private String name
"Dilip"
private float amount
1000.0
[2] : Account
private int acc_num
100103
private String name
"Tommy"
private float amount
10000.0
Local variables
String[] args = <object refere
Scanner sc = <object referer
Account[] aob = <object refe
int i = 3
Halt
Step
Step Into
Continue
Terminate